

Lab 2: Introduction to VHDL

The first step is to create an entity declaration before specifying the inputs and outputs. The interface is defined by the entity. The entity declaration gives the entity a name and specifies how it interacts with its surroundings. The "PORT" command is then used to define the inputs and outputs.¹ A port map connects signals in an architecture to ports on a specific instance. A block or a configuration can also contain port maps. Positional association or named association can be used to list the links.² Constraint files are simple files that can be compiled to improve the overall process intelligence.³ Each constraint group in a constraint document addresses one or more objects and has one or more constraints.³ The goal of designing a testbench is to give the design under test all conceivable inputs and view the results in an understandable way.

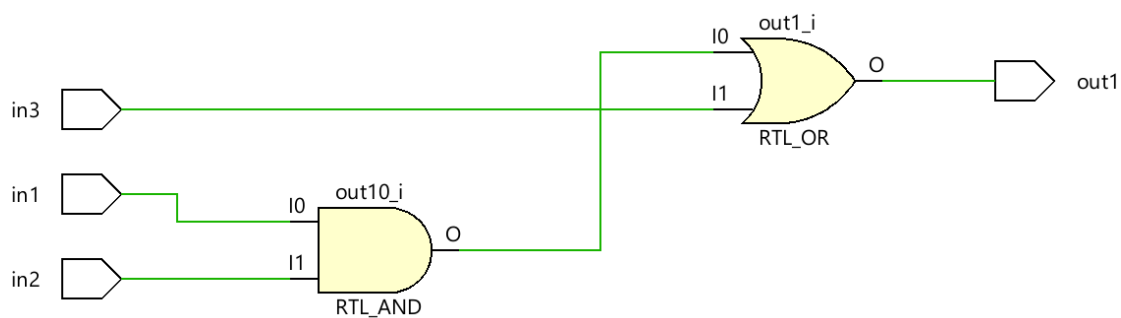


Figure 1: Schematic of the circuit

¹ https://web.engr.oregonstate.edu/~traylor/ece474/vhdl_lectures/entity_arch_ports.pdf

² https://www.hdlworks.com/hdl_corner/vhdl_ref/VHDLContents/PortMap.htm#:~:text=Description%3A,associa

³ <https://techdocs.altium.com/display/FPGA/How+it+Works+-+Configurations+and+Constraint+Files>

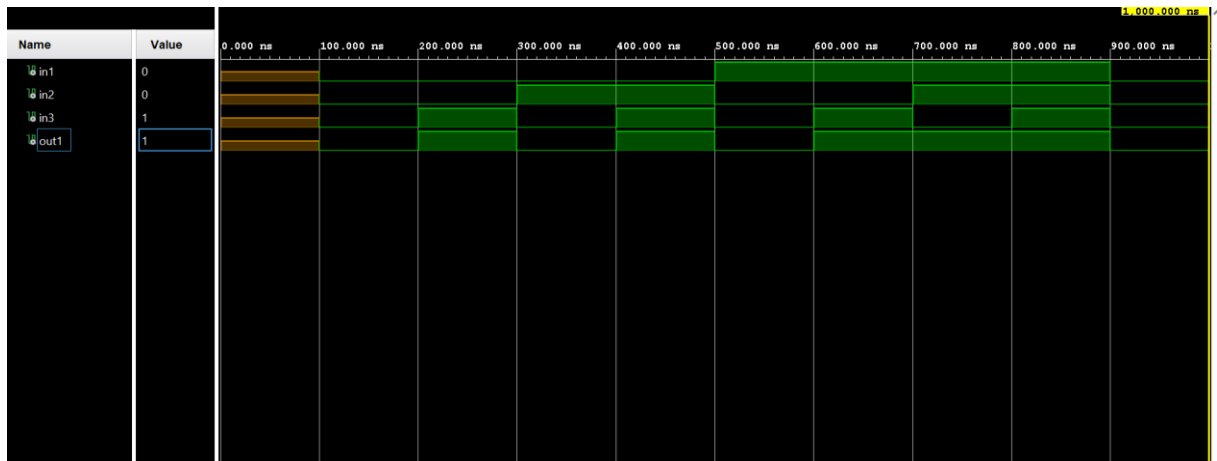


Figure 2: Testbench waveform

TOP MODULE

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity lab1 is
    Port ( in1 : in STD_LOGIC;
          in2 : in STD_LOGIC;
          in3 : in std_logic;
          out1 : out STD_LOGIC);
end test3;

architecture Behavioral of lab1 is
begin
    out1 <= (in1 and in2) or in3 ;
end Behavioral;
```

In1	In2	In3	Out1
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Table 1: truth table of the circuit

Testbench

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity testBench3 is
end testBench3;

architecture Behavioral of testBench3 is
COMPONENT lab1
PORT( in1 : IN STD_LOGIC;
in2 : IN STD_LOGIC;
in3 : IN std_logic;
out1 : OUT STD_LOGIC);
END COMPONENT;

SIGNAL in1 : STD_LOGIC;
SIGNAL in2 : STD_LOGIC;
SIGNAL in3 : std_logic;
SIGNAL out1 : STD_LOGIC;

BEGIN

    UUT: lab1 PORT MAP(
in1 => in1,
in2 => in2,
in3 => in3,
out1 => out1
);

    testBench3 : PROCESS
BEGIN
wait for 100 ns;

in1<='0';
in2<='0';
in3<='0';

wait for 100 ns;

in1<='0';
in2<='0';
```

```
in3<='1';

wait for 100 ns;

in1<='0';

in2<='1';

in3<='0';

wait for 100 ns;

in1<='0';
in2<='1';

in3<='1';wait for 100 ns;

in1<='1';
in2<='0';
in3<='0';

wait for 100 ns;

in1<='1';
in2<='0';
in3<='1';wait for 100 ns;

in1<='1';
in2<='1';
in3<='0';

wait for 100 ns;

in1<='1';
in2<='1';
in3<='1';

END PROCESS;

end Behavioral;
```

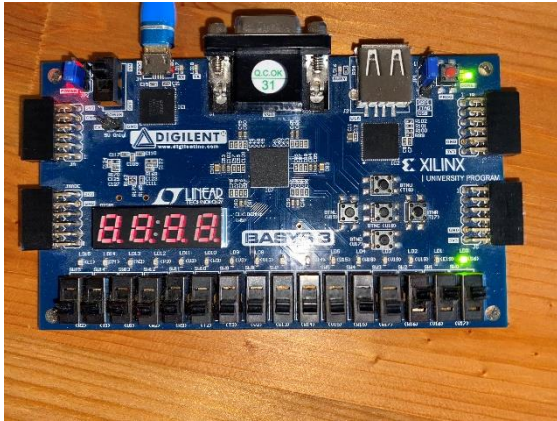


Figure 3: in1=1, in2=1, in3=1, out1=1

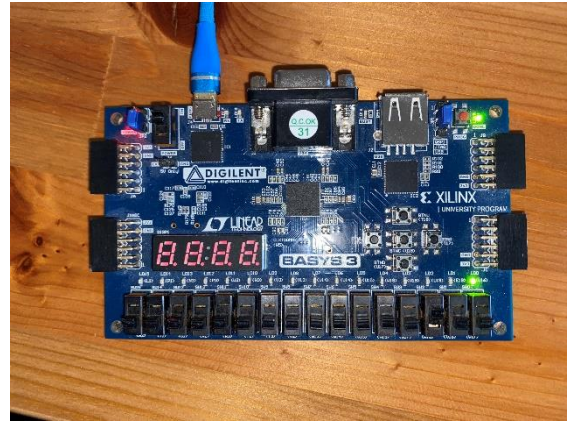


Figure 6: in1=1, in2=0, in3=0, out1=1

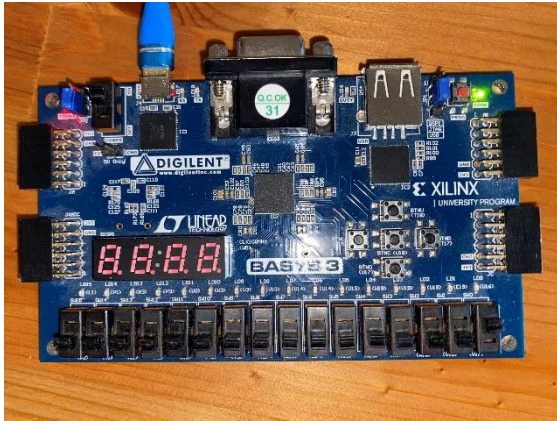


Figure 4: in1=1, in2=0, in3=0, out1=0

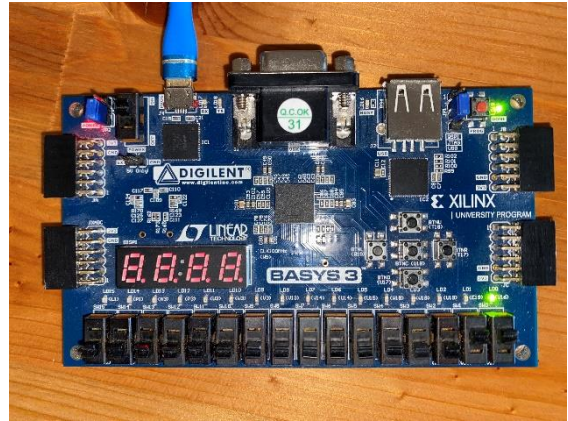


Figure 7: in1=0, in2=1, in3=1, out1=1

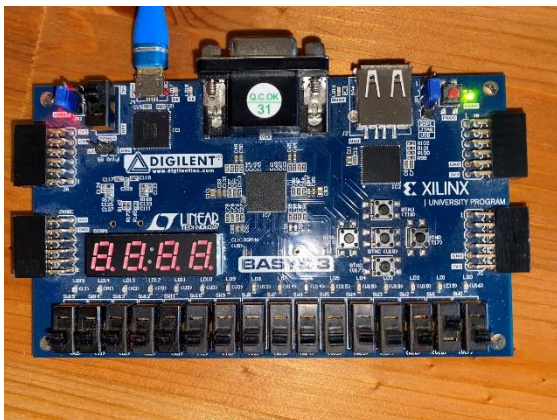


Figure 5: in1=0, in2=1, in3=0, out1=0